Chart & Institutional Patterns Reference Guide

A Technical Manual for Automating Strategy Lookback Windows

Executive Summary: Automating a trading strategy requires converting visual price formations into logical, mathematical rules. The core constraint when programming these patterns is defining a precise **Lookback Window**—the number of historical candles your code must evaluate in memory at any given moment to process arrays, confirm structure, and execute trades.

1. Retail & Classic Chart Patterns

Traditional chart patterns map the psychological battle between buyers and sellers. In algorithmic engines, these geometric shapes are parsed by tracking dynamic swing points (local peaks and troughs).

Reversal Formations

• **Double Top / Double Bottom:** Price tests a structural level twice with minor variance, failing to break out and reversing past the neckline. **Lookback: 40 - 80 candles**. *Logic:* Detect two peaks within a 5% price band separated by 10-20 candles.

• **Head and Shoulders (Standard/Inverted):** Three consecutive swing points where the center peak (head) is higher than the flanking shoulders, signaling exhaustion. **Lookback: 60 - 120 candles**. *Logic:* Track Peak 2 > Peak 1 and Peak 3 ≈ Peak 1.

• **Triple Top / Triple Bottom:** Three consecutive tests of an identical horizontal barrier followed by a powerful breakout. **Lookback: 50 - 100 candles**. *Logic:* Code an array filter for three local extremes hitting a tightly calibrated ceiling variance.

Continuation Formations

• **Bullish / Bearish Flags:** A sharp, near-vertical price spike (flagpole) followed by a tight, sloping consolidation channel against the trend. **Lookback: 20 - 50 candles**. *Logic:* Detect an aggressive X% move under 10 bars, then parallel channel limits.

• **Pennants:** A flagpole spike followed by a tiny, symmetrical converging consolidation triangle. **Lookback: 20 - 40 candles**. *Logic:* Measure rapid initial momentum followed by contracting lower highs and higher lows over 5-15 bars.

• **Cup and Handle:** A smooth, rounded U-shaped base followed by a shallow downward drift (the handle). **Lookback: 80 - 150+ candles**. *Logic:* High complexity. Requires fitting a quadratic curve or tracking sequential multi-bar arc segments.

Bilateral & Converging Formations

• **Symmetrical Triangles:** Two trendlines converge evenly (lower highs and higher lows), building pressure for a breakout. **Lookback: 40 - 90 candles**. *Logic:* Generate linear regression lines through swing points; check if lines intersect forward.

• **Ascending / Descending Triangles:** A flat horizontal barrier paired with a sloping trendline pushing price into it. **Lookback: 40 - 90 candles**. *Logic:* Horizontal support/resistance array matching combined with a clear linear slope parameter.

• **Rising / Falling Wedges:** Price tapers tightly within a channel moving in a single direction. Both boundaries slope together, but the trailing line is steeper. **Lookback: 50 - 100 candles**. *Logic:* Test for positive/negative directional slope synchronization with convergence.

2. Institutional & Smart Money Concepts (SMC)

Institutional systems ignore classic trendlines. Instead, they execute orders based on liquidity distribution, algorithmic market manipulation, and structural inefficiency rebalancing.

• **Fair Value Gaps (FVG) / Inefficiencies:** An aggressive, large-bodied candle creates a price imbalance, leaving a spatial void. **Lookback: Exactly 3 candles**. *Logic:* Continuous check across 3-bar arrays. Bullish FVG if Candle 1 High < Candle 3 Low; Bearish if Candle 1 Low > Candle 3 High.

• **Order Blocks (OB):** The final transactional zone where market makers stack inventory prior to a violent market move. **Lookback: 5 - 25 candles**. *Logic:* Locate the last oppositional bar immediately preceding a verified Break of Structure (BOS).

• **Liquidity Sweeps (Stop Hunts):** Price spikes aggressively outside an obvious retail structure to engineer liquidity, then instantly snaps back. **Lookback: 20 - 60 candles**. *Logic:* Condition where Current High > Highest High of lookback, but close prints within the range under heavy volume.

• **Market Structure Shift (MSS) vs. BOS:** BOS marks trend continuation; MSS confirms structural reversals by breaking the critical swing low/high responsible for the extreme. **Lookback: 30 - 70 candles**. *Logic:* Maintain continuous variables tracking confirmed swing points.

• **Mitigation & Breaker Blocks:** Failed Order Blocks. When an algorithm breaches an existing OB, that coordinate flips, transitioning into an inverted support/resistance zone. **Lookback: 40 - 100 candles**. *Logic:* Monitor inactive OB arrays; invert polarity once fully pierced by a candle close.

3. Algorithmic Matrix Checklist

This reference matrix details memory configuration and computing overhead when defining your trading scripts:

Pattern Category	Target Examples	Array Lookback Size	Computational Complexity
Micro Inefficiencies	FVGs, Engulfing, Doji	3 – 5 Candles	Very Low (Simple Inequalities)
Institutional Zones	Order Blocks, Sweeps, MSS	20 – 70 Candles	Medium (State Machine Tracking)
Macro Geometry	Head & Shoulders, Wedges, Cups	100 – 200+ Candles	High (Requires Peak/Trough Matrices)